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1- VACUUM SYSTEM THEORY

The Quiet Technology used on the Signa Twin*Speed* system is partially achieved by evacuating the air from around the Gradient coil and RF Body coil within the warm bore of the magnet. The components that make up the Vacuum system and are located in both the MR Equipment Room and the Magnet Room. See Illustration 1-1, Block Diagram.

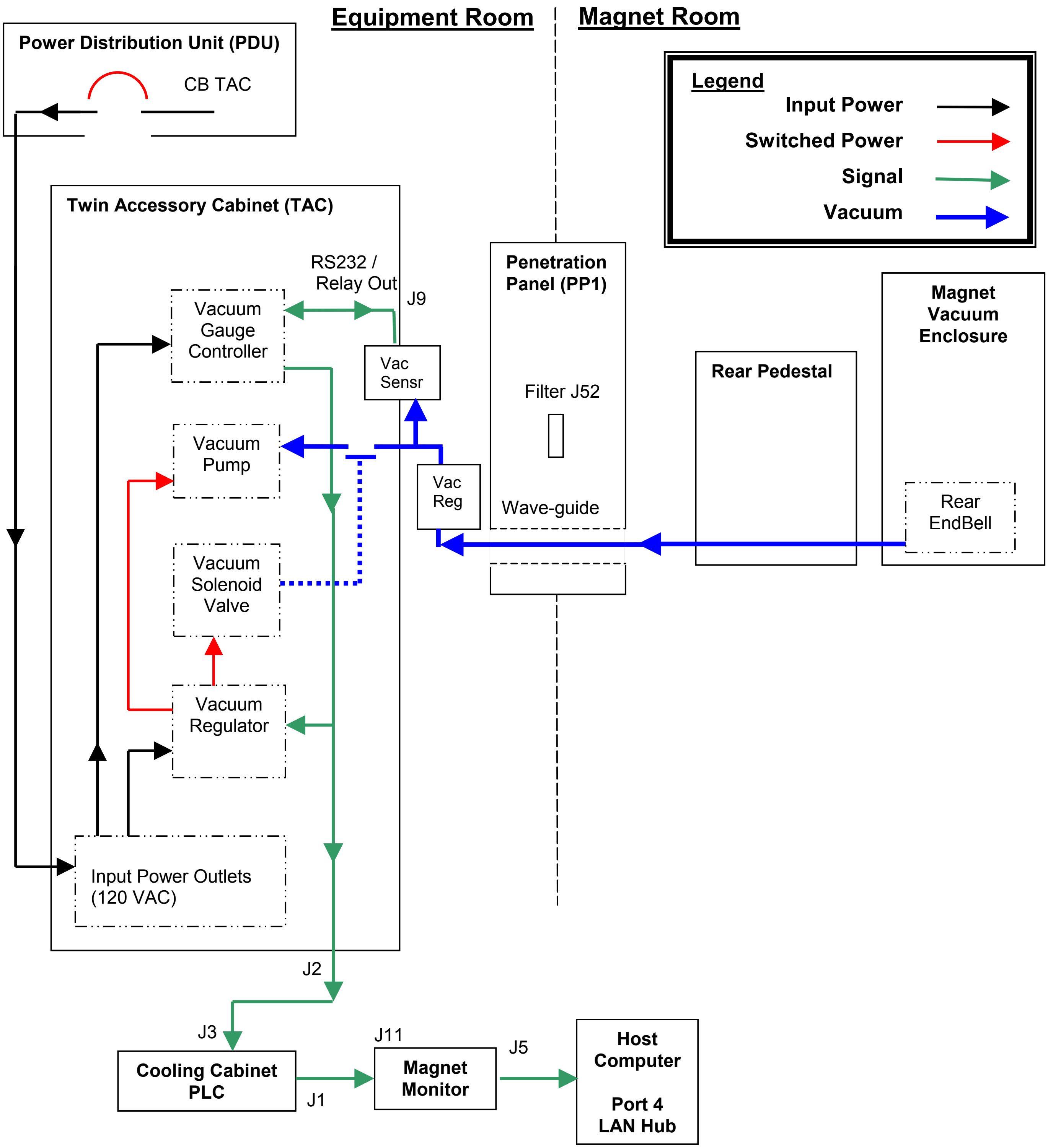

ILLUSTRATION 1-1
BLOCK DIAGRAM

Note

As of January, 2003 the software needed to communicate between the magnet monitor and the PLC has not been released. There is no method to remotely monitor the vacuum level.

Note

As of March, 2002 a new vacuum sensor and regulator replaced the existing units. The new configuration **does not require calibration**. See picture below to identify vacuum sensor.

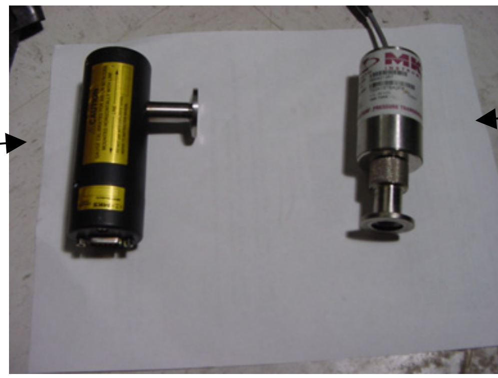

Illustration 1
Sensor Identification

Note

An FMI (**60631**) will be issued in March 2003 that will locate the Vacuum Sensor to the TAC cabinet. This FMI will also add a Vacuum Regulator to the vacuum line. The purpose of the Vacuum Regulator is to prevent the vacuum from going below 140 Torr if the primary Vacuum Control system fails.

On initial startup the Vacuum Gauge Controller will supply two 15-vdc signals to the Vacuum Regulator Box. One signal activates a solid-state relay, which supplies 120-vac to the Vacuum Pump, and the other 15-vdc signal activates a solid-state relay, which supplies 120-vac to the Vacuum Solenoid Valve. The Solenoid energizes causing the Vacuum Valve to open thereby allowing the Vacuum Pump to evacuate the Magnet Vacuum Enclosure. After about 10 minutes the vacuum level should decrease from atmosphere (760 torr at sea level) down to about 150 torr. Once the Gauge Controller reaches the lower limit of 150 torr (set point 1 - Low) it will remove the 15-vdc control signal for the Vacuum Solenoid Valve, allowing it to close. The Vacuum Pump continues to run with no load applied. The vacuum level within the Magnet Vacuum Enclosure will start to drift upward until the upper limit of 190 torr (set point 1 - High) is reached. The Gauge Controller re-activates the Vacuum Solenoid Valve causing it to open and the evacuation process is repeated. The typical drift time (no evacuation) for a normal operating system is 1 to 4 minutes.

The Mechanical Vacuum Regulator uses a flexible diaphram that will prevent the vacuum level from going below 140Torr. This regulator is factory set and is not field adjustable. The Vacuum Regulator was added as a safety feature via FMI (60631) to prevent body coil failures.

In the event of a failure where the vacuum level could be lower than 130 torr a second set point is used to de-active the Vacuum Pump. If the vacuum level ever reaches 130 torr (set point 2 – low) then the Gauge Controller will remove the 15-vdc control signal going to the solid-state relay that controls the input power to the Vacuum Pump. The control signal will remain at 0-vdc until the upper limit of 200 torr (set point 2 – High) is reached. The Gauge Controller will then change the control voltage from 0-vdc to 15-vdc activating the solid-state relay again. This feature was added to prevent damage to the RF Body Coil as components on the coil will arc and be damaged at vacuum levels less than 100 torr.

The Gauge Controller also sends the vacuum level data (analog format) to the Cooling Cabinet – Programmable Logic Control (PLC) where it is then relayed to the Magnet Monitor via the RS232 data link. The Magnet Monitor continuously monitors the data link and will send alarm status to the OLC and Host computer as configured by the operator.

2- TROUBLESHOOTING

CAUTION

Make sure the RF cabinet is powered off during vacuum troubleshooting. Damage to the RF Body Coil may occur with vacuum levels lower than 100 torr.

Make sure the power to the Gauge Controller Unit is OFF before disconnecting or connecting the Gauge (sensor) Cable. Connecting cables or loose cables with power applied can cause the gauge (sensor) to burn out.

2-1 Symptom/Probable Cause

Table 2-1 provides the symptom, probable cause and a checklist of items to aid in the troubleshooting process.

TABLE 2-1
VACUUM TROUBLESHOOTING

Symptom	**Probable Cause**	**Checklist**
Vacuum Pump not running	Loss of power or incorrect line voltage. (120 VAC +/- 10%)	Reset PDU – TAC Circuit breaker, must be ON.
		Reset TAC – Power Strip Circuit Breaker, must be ON.
		TAC – Vacuum Regulator Box Power cord is plugged into the Outlet Strip.
		Power cord wire continuity.
		Check fuse F1 in Vacuum Regulator Box.
		Check Voltage Indicator Moulding located next to the power switch for proper orientation. The correct voltage should line up with the arrow.
	Vacuum Pump Motor Thermal Overload	Ambient air temperature is over 40 ^{o} C. Supply Voltage too high. The fan is not working. Process gas too hot.
	Defective solid-state relay for Vacuum Pump	Check relay contacts
		Check Red LED illuminates with 15-vdc signal applied.
	Vacuum level too low, should be > 130 torr.	Check operation of Vacuum Valve. **Perform FMI 60631**

Symptom	Probable Cause	Checklist
	Defective Motor winding.	Check motor nominal DC resistance.
	Incorrect Power Switch Position	Check switch is in ON position.
Vacuum Pump too noisy	Worn bearings	Check motor current draw
	Pump is contaminated with solid particles	Remove Bell Housing and inspect scroll and tip seal.
Vacuum Valve does not cycle	Improper Gauge Controller Calibration. **Old style only**	Re-calibrate Gauge Controller and Sensor for atmosphere. See Appendix C –Altitude vs Barometric Pressure
	Incorrect Control set points. **Old style only**	Check set points See Appendix B – Supplier Hardware Information
	Degraded Vacuum Pump performance	Blocked inlet strainer Blocked exhaust line Gas ballast valve not closed Worn tip seal Worn bearings
	Defective solid-state relay for Vacuum solenoid valve.	Check solid-state relay for presence of 15-vdc signal.
	Leaks within the vacuum system hardware due to the following: Loose vacuum fittings. Damaged o-rings. Lack of vacuum grease on hose barb fittings or o-rings. Damaged sensor gauge. Damaged End Bell. Incorrect gasket thickness for RF tube. Damaged gasket – skewed. Loose screws on magnet interface ring. Gradient cable lead block skewed. Damaged vacuum hose.	Leak test all vacuum seals
Vacuum Gauge Controller Display Reading **HI**	Vacuum Gauge (sensor) is not connected to cable.	Check cable is connected to Vacuum Gauge (sensor).
	Pressure is higher than 995 torr. **Old style only.**	Check calibration of the Vacuum Gauge.
Vacuum Gauge Controller Display Reading **LO**.	Pressure is lower than –19 mTorr. **Old style only**.	Check calibration of the Vacuum Gauge.
		Vacuum Gauge should be located in the Rear Pedestal.

Symptom	Probable Cause	Checklist
Vacuum Gauge Controller Display Reading **OFF**.	Vacuum Gauge (sensor) cable disconnected from Vacuum Gauge Controller.	Check cable connection to the Vacuum Gauge Controller. See Appendix A – Cable Details
Vacuum Gauge Controller displays unreliable or noisy pressure readings.	Incorrect orientation of the Vacuum Gauge (sensor). **Old style only**	The Vacuum Gauge must be mounted with the inlet directed toward the floor and the Gauge body parallel to the floor.
	Faulty gauge cable or connection.	Tighten gauge cable connector screws. See Appendix A – Cable Details
	Contaminated or dirty gauge	Clean gauge following manufacturer's procedure.
	Gauge pins are shorted to the metal housing of the tube.	Use an ohmmeter to check gauge pins are open to ground.
Vacuum Gauge Controller readings seem wrong.	Contaminated or dirty gauge	Clean gauge following manufacturer's procedure.
	Defective Vacuum Gauge	Check for correct internal resistance values for the gauge. Make sure to follow the manufacturer's procedure.
Vacuum Pump does not shut off below 130 torr.	**FMI 60631 not completed**	**Complete FMI 60631**
	Vacuum Gauge Controller has incorrect set point values	Check set point configuration. See Appendix B – Supplier Hardware Information
	Faulty Vacuum Gauge Controller	Check 15-vdc control signal switches to 0-vdc.
	Vacuum Gauge Controller and Sensor not calibrated to Atmosphere	Re-calibrate hardware. See Appendix C – Altitude vs. Barometric Pressure
	Faulty Solid-state relay inside the Vacuum Control Box	120-vac output does not switch with a change in control voltage on the input.
	Faulty Cables	Check cables for proper continuity. See Appendix A – Cable Details

2-2 Troubleshooting Tips

Trying to locate a leak within the Twin*Speed* Magnet Enclosure vacuum system may be very tedious and time consuming. Here are some tips that should help to expedite the localization of the problem when troubleshooting poor vacuum performance issues.

1. Divide the system in half to determine if the problem is with the Vacuum Pump/Vacuum Valve/Vacuum Hose/Gauge Sensor or if the problem is in the Magnet Enclosure.
   a. Disconnect the vacuum hose from the Rear End Bell and cap it off using a KF25 Blank Flange. Start the Vacuum Pump and monitor the vacuum level. This will vary from site to site and is dependant on the length of vacuum hose installed on the system. If 100 ft of vacuum hose is used then the typical vacuum level should be about 5 torr and should take about 2 minutes. If this is not achieved then look for leaks on the vacuum hose fittings, vacuum gauge, vacuum solenoid valve. If these are OK then the Vacuum Pump has probably degraded and may need a tip seal replacement.
2. Make sure all o-rings and fittings have vacuum grease applied, especially the KF25 flange hose barb that is attached to the hose.
3. Make sure all vacuum clamps are tight.
4. Inspect o-rings for cracks or dirt. A fine thread lying across the face of the o-ring can be a major problem.
5. Check the screws that secure the Gradient Cable Block to the Rear End Bell. Leaks have occurred around the screws. Apply vacuum grease to the screw threads to seal the leak.

3- VACUUM LEAK TESTING

There are only two methods you should use when troubleshooting for leaks on a system that is under vacuum. Use either the Audible method or the Trace gas method. Never use liquids, as they will get sucked into the vacuum space, which could cause other problems.

3-1 Tools and Equipment Required

Item	Description	Part Number
1.	Helium gas cylinder 150 cu-ft (non-magnetic)	Order from Air Products
2.	Gas cylinder cart (non-magnetic)	46-258150P1
3.	Regulator kit with hose	46-306734G1
4.	Safety Glasses	46-271135P12
5.	Rubber Gloves	Field Supplied
6.	Vacuum grease	2297760
7.	Lint free cloth	Field Supplied
8.	Masking tape	Field Supplied
9.	Plastic bags	Field Supplied
10.	Non-magnet tool kit	46-320273G4
11.	Flashlight	Field Supplied
12.	Stopwatch	Field Supplied
13.	Digital Voltmeter	Field Supplied
14.	Extension cord	Field Supplied
15.	KF25 Blank Flange	46-235706P594
16.	KF16 Blank Flange	2294083-27

3-2 Safety

The following safety precautions must be followed when leak testing the system.

1. Always inspect the Pressure Regulator before installation onto the gas cylinder. Check for broken fittings or gauges. Do Not use if defects are found.
2. Always secure gas cylinders to either a cylinder cart or wall during use or storage.
3. Always wear personal protective equipment (safety glasses and gloves) during the set up, troubleshooting and tear down procedures.
4. Secure the area to unauthorized personnel where the work is being performed.
5. Follow established lock out / tag procedures prior to servicing equipment.

3-3 Audible Leak Test Method

In some cases the leak may be large enough for you to hear. Not all leaks sound the same as the orifice size and shape will affect the sound that is produced. The larger the leak … the lower the frequency or pitch of the sound.

1. Try to eliminate all ambient noise in the area where the equipment under test is located ie.
    a. Turn off the coldhead on the magnet.
    b. Turn off blowers/fans.
    c. Turn off music systems.
2. Next, turn off the vacuum system to get your ear familiar with the remaining sounds in the room.
3. Use a piece of tube or a stethoscope as a listening device to couple sounds to your ear.
4. Place the listening device near the piece of hardware that is suspect of leaking.

3-3 Trace Gas Method

IMPORTANT

The trace gas method of leak troubleshooting will only work for the older style Gauge controller. It will not work with the new style.

The Trace gas method is used when the leaks are not audible and difficult to find. Instruments with a high sensitivity to a specific trace gas are used to locate vacuum leaks of this type. The Vacuum Sensor and Gauge Controller used on the TwinSpeed system posses this feature and will react to the presence of helium when it is used as a trace gas. The following steps should be used to minimize false or erratic readings.

1. Make sure the display on the Gauge Controller is visible to the person administering the trace gas, or have someone watch the display and call out if the reading increases. The gauge controller can be removed from the Twin Accessory Cabinet and taken into the magnet room for troubleshooting purposes; it has a universal power supply so it can be powered from a room outlet if you have the correct power cord adaptor.
2. Administer the trace gas sparingly.
    a. Place the nozzle or end of the Tygon tubing near your cheek.
    b. Adjust the gas regulator so the output gas is just barely felt on your check.
3. Always start at the highest region on the hardware to be tested as helium rises which could result in a false reading in correlation to the area you are testing.
4. Isolate the area to be tested by taping a plastic bag over it and injecting the bag with a very small amount of helium gas. Then monitor the Gauge Controller and watch for an increase in the reading. The response should be within 30 seconds after the administration of the trace gas.

APPENDIX A – CABLE DETAILS

OLD STYLE Vacuum Sensor Cable

15 Pin Sub D Male Vacuum Gauge Controller	***9 Pin Sub D Female HPS/MKS 317 Sensor Tube***
Pin Number	**Pin Number**
2	Sensor (1) pins 2, 4, 5 & 9
3	Sensor (1) pins 2, 4, 5 & 9
4	Sensor (1) pins 2, 4, 5 & 9
5	Sensor (1) pin 7
6	Sensor (1) pin 8
7	Sensor (1) pins 1, 3 & 6
8	Sensor (1) pins 1, 3 & 6
1, 9, 10, 11, 12, 13, 14, and 15 – No Connection	

NEW STYLE Vacuum Sensor Cable

9 Pin Sub D Male Vacuum Gauge Controller	***9 Pin Sub D Female HPS/MKS 317 Sensor Tube***
Pin Number	**Pin Number**
1	Tube #1, Pin 2/Signal Output
2	
3	
4	Tube #1, Pin 7/ +15 vdc
5	
6	Tube #1, Pin 6/ -15 vdc
7	
8	Tube #1 pin 12/Signal Common
9	Tube #1 Pins 5/Pwr Comm & 15/Chassis control

Y-Adaptor Cable

15 Pin Sub D Male Vacuum Gauge Controller		*(2) 9 Pin Sub D cables – 9 Pin Male - Solid-State Relay Control 9 Pin Female - Analog Out*
Pin Number	**Signal Name**	**Pin Number**
1, 2, 4, 5, 7, 8, 10, 11, 12, 14, 15 – no connection		
3	Set point 1 control signal out (0 or 15 vdc)	Pin 3 - Male Connector
6	Set point 2 control signal out (0 or 15 vdc)	Pin 6 – Male Connector
9	Ground	Pins 8, 9 – Male Connector Pin 9 - Female Connector
13	Analog Out – 0 to 4.0 vdc	Pin 6 – Female Connector

APPENDIX B – SUPPLIER HARDWARE INFORMATION

BOC Edwards Dry Vacuum Pump Model XDS10-C

Parameter	Specification
Input Voltage	120 VAC +/- 10 % @ 50/60 hz +/- 3 hz
Nominal Current	60 Hz is 6.4 amps / 50 hz is 7.2 amps
Motor Winding Resistance	Hot to Neutral 1.9 ohms + / - 0.5 ohms; infinite resistance to ground.
Pumping Speed	9.3 cubic meters / hour @ 50 hz 11.1 cubic meters / hour @ 60 hz

Terranova Scientific Model 926-GE Dual Convection Gauge Controller. Note the new style gauge controler, model 908A does not require calibration. See Illustration 1. to identify sensor type.

Parameter	Specification
Input Voltage	120 vac; powered from PDU (see note 1)
Units of Display	Torr or mBar; user selectable
Display Resolution	Varies; from 0.1 mTorr below 10 mTorr, to 5 Torr above 100 Torr.
Set Points	Set Point 1 High & Low, configurable by the operator Set Point 2 High & Low, configurable by the operator **TwinSpeed Set Point Values** Set point 1 Low = 150 torr Set point 1 High = 190 torr Set point 2 Low = 130 torr Set point 2 High = 200 torr
Memory	Nonvolatile for VAC, ATM and Set Points
Outputs	**RS232** – Allows user to read pressure and configure set points; 9600 baud, 8-N-1; available through the D15 accessory connector. **Analog Output** – Calibrated, 12 bit resolution, logarithmic, 0.50 volts/decade; 0.0 mTorr = 0 volts; 10 mTorr = 0.50 volts; 100 mTorr = 1.00 volt etc., available through the D15 accessory connector. **Control Output** – The Model 926-GE is specially designed for GE. The outputs are (two) switched 15 vdc sources used to control external solid-state relays. See Note 2.
Note 1: The Model 926 incorporates a universal power supply which allows it to operate on any input voltage from 90 vac to 265 vac, 47 to 65 Hz.	
Note 2: The manufacturer has included an addendum to the Instruction Manual Terranova Scientific Model 926-GE Dual Convection Gauge Controller that covers these specific changes.	

MKS/HPS Type 317 Convection Tube (Vacuum Gauge Sensor). Note the new vacuum sensor is not sensitive to mounting orientation.

Parameter	**Specification**
Mounting Orientation	Gauge tube axis horizontal and inlet port pointing down.

Vacuum Valve Solenoid

Parameter	**Specification**
Input Voltage	120 VAC +/- 10% @ 50/60 Hz +/- 3 Hz
Nominal Current	0.150 amps
Nominal Solenoid Resistance	Nominal resistance 115.7 ohms
Valve Position	Normally closed with no power applied

Solid-State Relay

Parameter	**Specification**
Signal Input	+ 5 to + 15 vdc
Max Output Load Current Rating	20 amps for Vacuum Pump solid-state relay 10 amps for Vacuum Solenoid Valve solid-state relay
Visual Indicator	Red LED illuminates with presence of input signal.

APPENDIX C – ALTITUDE VS BAROMETRIC PRESSURE

ALTITUDE ABOVE SEA LEVEL		BAROMETRIC PRESSURE
FEET	METERS	TORR
-1000	-305	790
-500	-153	775
0	0	760
500	153	745
1000	305	735
1500	458	720
2000	610	705
2500	763	695
3000	915	680
3500	1068	670
4000	1220	655
4500	1373	645
5000	1526	630
6000	1831	610
7000	2136	585
8000	2441	565
9000	2746	545
10000	3050	520
15000	4577	430
20000	6102	350

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Nov 28, 2001	P. Senski	Initial Draft of Troubleshooting Procedure
0	Dec 4, 2001	P. Senski	Initial Release – incorporated changes from internal review
1	Feb 12, 2003	Kargard	Changes for new style vacuum gauge and sensor, and vacuum regulator